# Configuring AAA

Difficulty (out of 3 stars): ★★★ Recommended study time: 2 hours

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# 1 About This Lab

## 1.1 Task Description

Authentication, authorization, and accounting (AAA) provides a management mechanism for network security.

AAA provides the following functions:

- Authentication: verifies whether users are allowed to access the network.
- Authorization: authorizes users to access specific services.
- Accounting: records the network resources used by users for charging purposes.

Users can use one or more security services provided by AAA. For example, if a company wants to authenticate employees that access certain network resources, the network administrator only needs to configure an authentication server. If the company also wants to record the operations performed by employees on the network, an accounting server is needed.

In summary, AAA authorizes users to access specific resources and records user operations. AAA is widely used because it features good scalability and facilitates centralized user information management. AAA can be implemented using multiple protocols. RADIUS is most commonly used in actual scenarios.

After completing this task, you will gain a clear understanding of local AAA and learn how to configure local AAA to manage and control resources for remote Telnet users.

## 1.2 Objectives

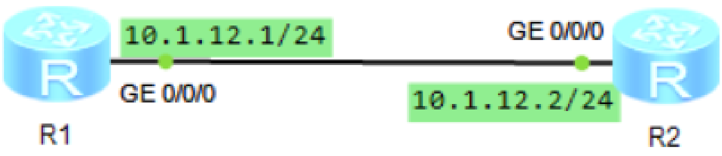
On completing this task, you will be able to:

1. Configure local AAA authentication and authorization schemes.
2. Create a domain.
3. Create a local user.
4. Understand domain-based user management.

# 2 Task Preparation

## 2.1 Network Topology

**Figure 2-1** Lab topology for local AAA configuration



R1 functions as a client, and R2 functions as a network device. Access to the resources on R2 needs to be controlled so that only authenticated users can access specific resources. Therefore, you need to configure local AAA authentication on R1 and R2 and manage users based on domains, and configure the privilege level for authenticated users.

## 2.2 Initial Configuration

- Initial configuration of R1

```
<Huawei>system-view
Enter system view, return user view with Ctrl+Z.
[Huawei]sysname R1
[R1]int g0/0/0
[R1-GigabitEthernet0/0/0]ip address 10.1.12.1 24
[R1-GigabitEthernet0/0/0]quit
[R1]
```

- Initial configuration of R2

```
<Huawei>system-view
Enter system view, return user view with Ctrl+Z.
[Huawei]sysname R2
[R2]int g0/0/0
[R2-GigabitEthernet0/0/0]ip address 10.1.12.2 24
[R2-GigabitEthernet0/0/0]quit
[R2]
```

# 3 Task Implementation

## 3.1 Checking Basic Configurations

Omitted

## 3.2 Configuring AAA Schemes

Configure authentication and authorization schemes.

```
[R2-aaa]aaa
```

Enter the AAA view.

```
[R2-aaa]authentication-scheme datacom
Info: Create a new authentication scheme.
```

Create an authentication scheme named **datacom**.

```
[R2-aaa-authen-datacom]authentication-mode local
```

Set the authentication mode of the authentication scheme to local authentication.

```
[R2-aaa-authen-datacom]quit
[R2-aaa]authorization-scheme datacom
Info: Create a new authorization scheme.
```

Create an authorization scheme named **datacom**.

```
[R2-aaa-author-datacom]authorization-mode local
```

Set the authorization mode of the authorization scheme to local authorization.

```
[R2-aaa-author-datacom]quit
```

A device functioning as an AAA server is called a local AAA server, which performs user authentication and authorization, but not user accounting.

Similar to the remote AAA server, the local AAA server need to have the user names, passwords, and permission information of local users configured. A local AAA server performs authentication and authorization faster than a remote AAA server, which reduces operation costs. However, the information storage capacity of a local AAA server is limited by the available space on the device hardware.

## 3.3 Creating a Domain and Applying AAA Schemes to the Domain

```
[R2]aaa
[R2-aaa]domain datacom
```

The device manages users based on domains. A domain is a group of users and each user belongs to a domain. A user uses the AAA configuration in the domain to which the user belongs. Create a domain named **datacom**.

```
[R2-aaa-domain-datacom]authentication-scheme datacom
```

Specify the authentication scheme named **datacom** for users in the domain.

```
[R2-aaa-domain-datacom]authorization-scheme datacom
```

Specify the authorization scheme named **datacom** for users in the domain.

## 3.4 Configuring a Local User

1. Create a local user and set a password for the user.

   ```
   [R2-aaa]local-user user01 password cipher Huawei@123
   Info: Add a new user.
   [R2-aaa]
   ```

   If the user name contains a delimiter of an at sign (@), the character string before the at sign is the user name and that following the at sign is the domain name. If the user name does not contain @, the entire character string is the user name and the domain name is the default one.

2. Configure parameters for the local user, such as access type and privilege level.

   ```
   [R2-aaa]local-user user01 service-type telnet
   [R2-aaa]
   ```

   The **local-user service-type** command sets the access type for a local user. After you specify the access type for a user, the user can successfully log in only through the configured access type. If the access type is set to **telnet**, the user cannot access the device through a web page. Multiple access types can be configured for a user.

   ```
   [R2-aaa]local-user user01 privilege level 3
   [R2-aaa]
   ```

   Specify a privilege level for the local user. After logging in to the device, a user can run only the commands of the same privilege level or lower privilege levels.

## 3.5 Enabling the Telnet Function on R2

Enable the Telnet server function on the device. If the function has been enabled, an error may be reported when you enable it. This function is enabled on some devices by default.

```
[R2]telnet server enable
Error: TELNET server has been enabled
[R2]user-interface vty 0 4
[R2-ui-vty0-4]authentication-mode aaa
[R2-ui-vty0-4]quit
[R2]
```

The **authentication-mode** command configures an authentication mode for accessing the user interface. By default, the authentication mode for accessing the user interface is not configured. An authentication mode must be configured for accessing the user interface. Otherwise, users cannot log in to the device.

## 3.6 Verifying the Configuration

1. Telnet to R2 from R1.

   The following output shows that R1 has logged in to R2.

```
<R1>telnet 10.1.12.2
  Press CTRL_] to quit telnet mode
  Trying 10.1.12.2 ...
  Connected to 10.1.12.2 ...

Login authentication


Username:user01
Password:
<R2>
<R2>
<R2>quit

  Configuration console exit, please retry to log on

  The connection was closed by the remote host
<R1>
```

2. Check the logged-in user on R2.

```

[R2]display users
  User-Intf      Delay      Type     Network Address       AuthenStatus      AuthorcmdFlag
+ 0     CON 0     00:00:00                                       pass
  Username : Unspecified

  129 VTY 0     00:00:17   TEL      10.1.12.1                   pass
  Username : user01


[R2]
```

# 4 Verification

Omitted

# 5 Quiz

1. What authentication, authorization, and accounting modes are supported by AAA?

   AAA supports the following authentication modes: non-authentication, local authentication, and remote authentication. AAA supports the following authorization modes: non-authorization, local authorization, and remote authorization. AAA supports two accounting modes: non-accounting and remote accounting.

2. When a new common user is configured with local authentication but is not associated with a user-defined domain, which domain does the user belong to?

   If the domain to which a user belongs is not specified when the user is being created, the user is automatically associated with the default domain named **default** (administrators are associated with the **default_admin** domain).